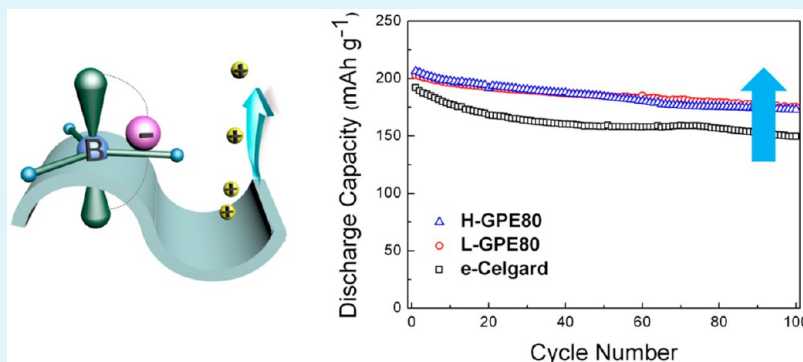


Gel Polymer Electrolytes Containing Anion-Trapping Boron Moieties for Lithium-Ion Battery Applications

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S Supporting Information



ABSTRACT: Gel polymer electrolytes (GPEs) based on semi-interpenetrating polymer network (IPN) structure for lithium-ion batteries were prepared by mixing boron-containing cross-linker (BC) composed of ethylene oxide (EO) chains, cross-linkable methacrylate group, and anion-trapping boron moiety with poly(vinylidene fluoride) (PVDF) followed by ultraviolet light-induced curing process. Various physical and electrochemical properties of the GPEs were systematically investigated by varying the EO chain length and boron content. Dimensional stability at high temperature without thermal shrinkage, if any, was observed due to the presence of thermally stable PVDF in the GPEs. GPE having 80 wt % of BC and 20 wt % of PVDF exhibited an ionic conductivity of 4.2 mS cm^{-1} at 30°C which is 1 order of magnitude larger than that of the liquid electrolyte system containing the commercial Celgard separator (0.4 mS cm^{-1}) owing to the facile electrolyte uptake ability of EO chain and anion-trapping ability of the boron moiety. As a result, the lithium-ion battery cell prepared using the GPE with BC showed an excellent cycle performance at 1.0 C maintaining 87% of capacity during 100 cycles.

KEYWORDS: lithium-ion battery, gel polymer electrolyte, semi-IPN, boron, lithium transference number

INTRODUCTION

Lithium-ion batteries (LIBs) widely used in energy conversion and storage devices are the most promising power sources for various applications such as portable electronics, electric vehicles, and large-capacity electrical energy storage system.^{1,2} Because liquid electrolytes based on organic solvents possess safety issues such as flammability or leakage, polymer electrolytes have been intensively studied to substitute the conventional separator/liquid-based electrolyte system.^{3–10} Although solid polymer electrolytes (SPEs) have been suggested as a breakthrough for resolving the safety issues of liquid electrolytes by removing all the volatile liquid components, utilization of the SPEs have been often limited due to their low ionic conductivity originated from the intrinsically small mobility of solid polymer chains and large interfacial resistance.^{5–13}

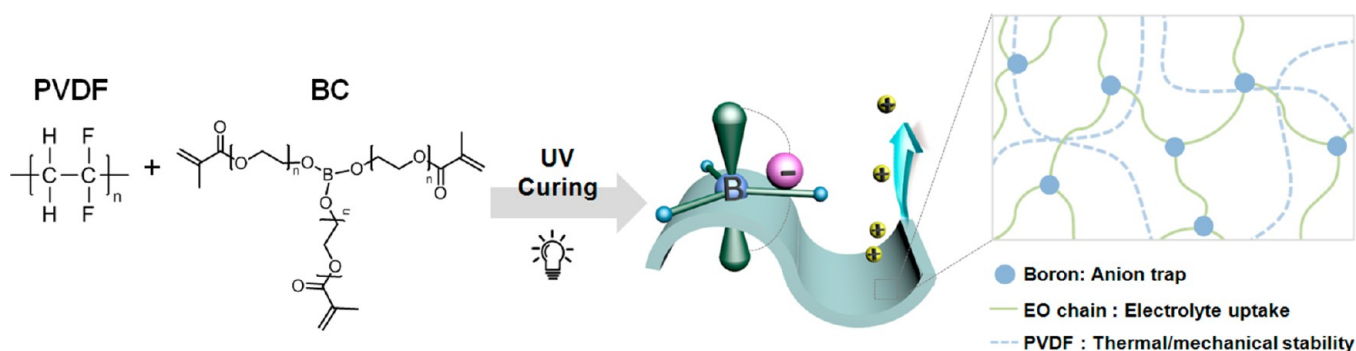
Gel polymer electrolytes (GPEs) comprising swollen polymer matrix by liquid electrolyte combine the advantages of both liquid electrolytes and SPEs such as reliable safety and reasonably high ionic conductivity compared to the liquid

electrolyte systems.^{14–16} Poly(methyl methacrylate) (PMMA),^{17,18} poly(acrylonitrile) (PAN),^{19–21} poly(vinylidene fluoride) (PVDF),^{22,23} and poly(ethylene oxide) (PEO)^{18,21,24,25} have been suggested as polymer matrices for the GPEs, because the functional groups in these polymers interact with lithium ions and liquid electrolytes. Among them, PVDF and PEO derivatives have been extensively studied for the matrix of GPEs because PVDF has good mechanical/electrochemical stability with a high dielectric constant, and PEO derivatives can absorb a large amount of liquid electrolyte. However, it is well-known that PVDF cannot hold enough of the liquid electrolyte because of its semicrystalline structure,²⁶ and the PEO derivatives are not mechanically stable enough to maintain the free-standing film state because they are easily swollen by liquid electrolytes.^{21,27} Therefore, studies combining the advantages of PVDF and PEO as well as minimizing their

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Scheme 1. Schematic Illustration of GPE System with a Semi-Interpenetrating (IPN) Structure Having PVDF and Boron-Containing Crosslinker (BC)



disadvantages for obtaining GPEs with high ionic conductivity and mechanical stability are in progress.^{26,28}

The lithium transference number (t_{Li^+}) defined as a fraction of the total current carried in an electrolyte by a lithium ion is a critical parameter to obtain good cycle performance of LIBs.²⁹ Although the t_{Li^+} value is very desirable to be close to unity, most electrolytes have t_{Li^+} values smaller than 0.5 because of the concentration polarization of ions during the repeated charge/discharge cycles, resulting in the deterioration of long-term cycle performance.³⁰ The t_{Li^+} values of LIB could be increased by introducing single-ion conductor,^{31,32} ceramic fillers such as SiO_2 , Al_2O_3 , and TiO_2 ,^{33,34} and Lewis acids to the electrolytes.³⁵ Especially boron moieties in LIB systems can improve the electrochemical properties because they effectively trap anions in the lithium salts through Lewis acid–base interaction between the vacant p orbital of the boron and basic anions of the lithium salt.^{36–40,60–64}

In this study, GPEs based on semi-interpenetrating polymer network (IPN) structure containing PVDF and cross-linkers having anion-trapping boron moieties and ethylene oxide (EO) chains were prepared (Scheme 1). Various physical and electrochemical properties were systematically investigated by varying the EO chain length and the content of boron-containing cross-linkers (BCs). Our cross-linked GPE system based on PVDF for the structural matrix, EO chain for electrolyte uptake, and boron for anion-trapping unit showed a high ionic conductivity of 4.2 mS cm^{-1} at 30°C and large lithium transference number of 0.82, resulting in excellent cycle performance.

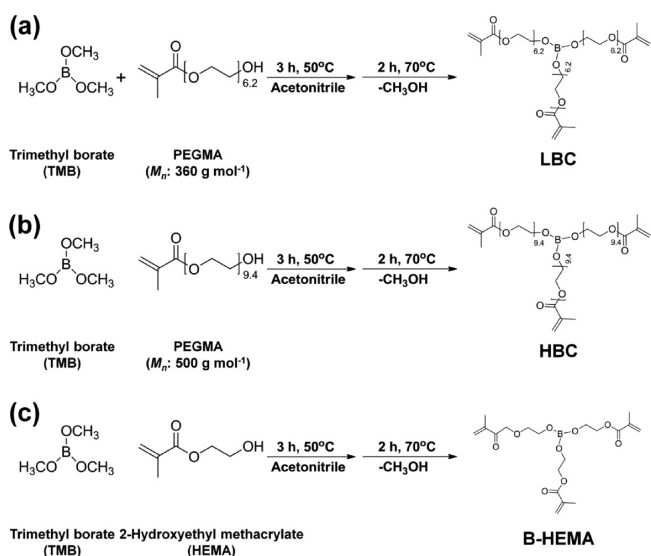
EXPERIMENTAL SECTION

Chemicals and Materials. Poly(ethylene glycol) methacrylate (PEGMA, average $M_n = 360 \text{ g mol}^{-1}$ and 500 g mol^{-1}), trimethyl borate (TMB, $\geq 99.8\%$), poly(vinylidene fluoride) (PVDF, average $M_w = 534\,000 \text{ g mol}^{-1}$), 2-hydroxyethyl methacrylate (HEMA), trimethylolpropane ethoxylate triacrylate (ETPTA, average $M_n = 912 \text{ g mol}^{-1}$), 2-hydroxy-2-methyl-1-phenyl-1-propanone (HMPP), and *N*-methyl-2-pyrrolidone (NMP, 99.5%) were purchased from Aldrich and used as received. Acetonitrile was purchased from TCI and stored over molecular sieves prior to use. 1.0 M lithium bis(trifluoromethane sulfonyl) imide (LiTFSI) electrolyte solution in a mixture of ethylene carbonate (EC) and diethylene carbonate (DEC) (1:1 vol %) was purchased from PANAXETEC. Co., Ltd. All other reagents and solvents were obtained from reliable commercial sources and used as received.

Synthesis of Boron-Containing Cross-Linkers (BCs) Such as LBC, HBC, and B-HEMA. A series of boron-containing cross-linkers (BCs) were synthesized via substitution reaction using two types of poly(ethylene glycol) methacrylate (PEGMA) and 2-hydroxyethyl

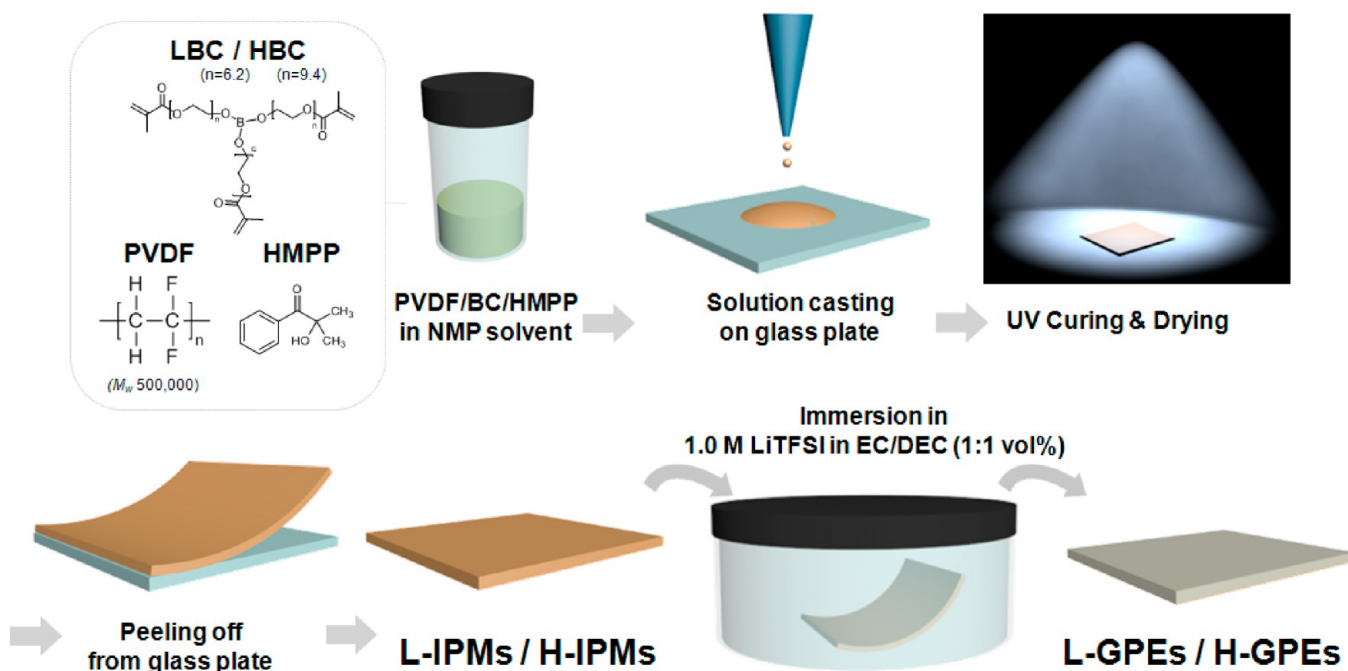
methacrylate (HEMA) with trimethyl borate (TMB). For convenience, BCs prepared from PEGMA with M_n of 360 and 500 g mol^{-1} are abbreviated as LBC and HBC, respectively, where L and H represent lower and higher molecular weight, respectively, and BC prepared from HEMA is abbreviated as B-HEMA. Their chemical structures are shown in Scheme 2. For the preparation of LBC, TMB

Scheme 2. Synthesis of BCs Such as (a) LBC, (b) HBC, and (c) B-HEMA, where LBC and HBC Are Prepared from PEGMAs with M_n of 360 and 500 g mol^{-1} , Respectively



(5.2 mL, 4.7 mmol) and PEGMA ($M_n = 360 \text{ g mol}^{-1}$, 5.0 g, 14 mmol) were dissolved in 50 mL of anhydrous acetonitrile, and the solution was stirred at 50°C for 3 h to equilibrate the reaction in dry nitrogen atmosphere. Then, the reaction temperature was raised to 70°C and stirred for another 2 h under nitrogen purging to eliminate methanol that generated as a byproduct. Unreacted TMB and residual solvent were removed under reduced pressure at 50°C and dried under high vacuum condition at room temperature for 2 days to produce 4.4 g of LBC in 87% of yield. HBC was also prepared by the same procedure except using PEGMA with M_n of 500 g mol^{-1} in 83% of yield. BC having HEMA groups (B-HEMA) without EO chains was also prepared from TMB and 2-hydroxyethyl methacrylate (HEMA) by the same procedure. Since B–O bond has been known to be vulnerable to a small amount of water in the atmosphere, BCs were stored in an Ar-filled glovebox to prevent possible hydrolysis.⁴¹ ^1H NMR [300 MHz, $CDCl_3$, δ (ppm), TMS ref] of LBC and HBC: 6.13 (vinyl, CH), 5.58 (vinyl, CH), 4.31 ($CH_2\text{--O--C(O)}$), 4.05–3.88 ($CH_2\text{--O--B}$), 3.57–3.77 ($CH_2\text{--CH}_2\text{--O}$), 1.95 (isobutyl, CH_3). ^1H NMR [300 MHz, $CDCl_3$, δ (ppm), TMS ref] of B-HEMA: 6.11 (vinyl, CH), 5.56 (vinyl, CH), 4.24 ($CH_2\text{--O--C(O)}$), 3.91 ($CH_2\text{--O--B}$), 1.96 (isobutyl, CH_3).

Scheme 3. Preparation of Semi-IPN Polymer Membranes (IPMs) and Gel Polymer Electrolytes (GPEs)



Scheme 4. Compositions of Semi-IPN Polymer Membranes (IPMs) and Gel Polymer Electrolytes (GPEs); (a) L-IPMs/L-GPEs, (b) H-IPMs/H-GPEs, (c) BH-IPM20/BH-GPE20, (d) E-IPM80/E-GPE20, and (e) PVDF/e-PVDF and Celgard/e-Celgard

(a)

LBC
(n=6.2)

PVDF

UV Curing

Immersion in liquid electrolyte

L-IPMs

L-GPEs

20 wt%	80 wt%	L-IPM20	L-GPE20
40 wt%	60 wt%	L-IPM40	L-GPE40
60 wt%	40 wt%	L-IPM60	L-GPE60
80 wt%	20 wt%	L-IPM80	L-GPE80

(b)

HBC
(n=9.4)

PVDF

UV Curing

Immersion in liquid electrolyte

H-IPMs

H-GPEs

20 wt%	80 wt%	H-IPM20	H-GPE20
40 wt%	60 wt%	H-IPM40	H-GPE40
60 wt%	40 wt%	H-IPM60	H-GPE60
80 wt%	20 wt%	H-IPM80	H-GPE80

(c)

B-HEMA

PVDF

UV Curing

Immersion in liquid electrolyte

BH-IPM

BH-GPE

20 wt%	80 wt%	BH-IPM20	BH-GPE20
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(d)

ETPTA

PVDF

UV Curing

Immersion in liquid electrolyte

E-IPM

E-GPE

80 wt%	20 wt%	E-IPM80	E-GPE80
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(e)

PVDF

Celgard

Immersion in liquid electrolyte

Immersion in liquid electrolyte

e-PVDF

e-Celgard

Preparation of Semi-IPN Polymer Membranes (IPMs). The semi-interpenetrating polymer network (IPN) polymer membranes (IPMs) containing BCs (LBC or HBC) and PVDF with different weight ratios such as 20:80, 40:60, 60:40, and 80:20 were prepared by a solution casting method followed by ultraviolet (UV) light-induced curing process. The schematic illustration of the preparation of IPMs is shown in Scheme 3. BC (LBC or HBC), PVDF, and HMPP were dissolved in NMP and the homogeneous solutions were obtained, where 10 wt % of HMPP to BC was used. The solution was cast onto a glass plate, and then it was exposed to UV light (OV-11 ultraviolet lamp, 60 Hz, FORCELAMP Co., LTD, Korea) for 5 min to form a semi-IPN structure. The residual solvent was evaporated at 85 °C on a hot plate and further dried at 60 °C under high vacuum for several days. The obtained film was peeled off from the glass plate, and then placed in a high vacuum condition for a week at 60 °C. The thickness of IPMs could be controlled by varying the amount of BC and PVDF.

For convenience, IPMs prepared from 20, 40, 60, and 80 wt % of LBC were named as L-IPM20, L-IPM40, L-IPM60, and L-IPM80, respectively, and those from 20, 40, 60, and 80 wt % of HBC were named as H-IPM20, H-IPM40, H-IPM60, and H-IPM80, respectively (Scheme 4). For morphological and electrochemical studies, PVDF film was prepared by using the solution casting method. PVDF powder was dissolved in NMP and cast onto a glass plate. After removing the solvent, the PVDF film was peeled off from the plate and the film was dried under high vacuum condition at 60 °C. UV-cured LBC and HBC films were also prepared from the same procedures used for the preparation of L-IPMs and H-IPMs.

Preparation of Gel Polymer Electrolytes (GPEs). The gel polymer electrolytes (GPEs) were prepared by immersing the IPMs into 1.0 M LiTFSI in EC:DEC (1:1 vol %) for 24 h. The excessive liquid electrolyte residue on the membrane surface was carefully wiped off with a tissue. The GPEs from L-IPMs and H-IPMs are abbreviated

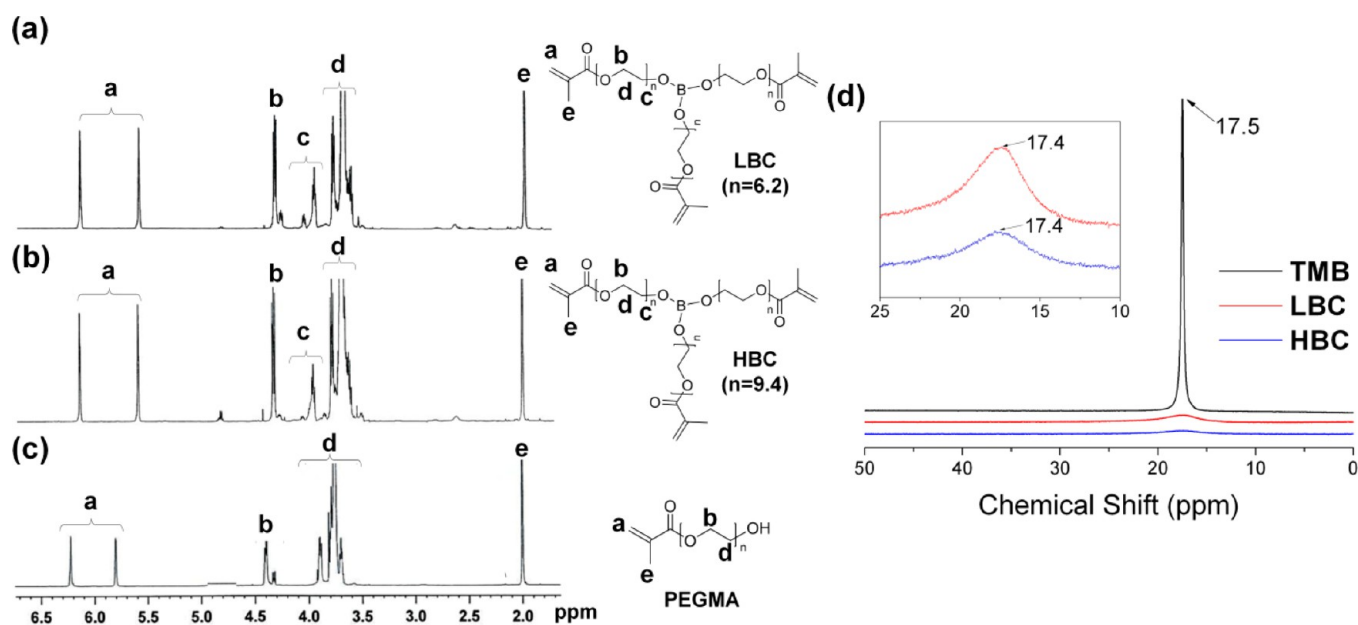


Figure 1. ^1H NMR spectra of (a) LBC, (b) HBC, and (c) PEGMA ($M_n = 360 \text{ g mol}^{-1}$) and (d) ^{11}B NMR spectra of TMB, LBC, and HBC.

as L-GPEs and H-GPEs, respectively. The numbers after the abbreviations represent the weight percent of LBC or HBC. The GPEs without EO chains and boron moieties were prepared from B-HEMA and ETPTA instead of LBC or HBC, respectively. The GPE based on B-HEMA and ETPTA are abbreviated as BH-GPE and E-GPE, respectively. The numbers after the abbreviations represent the weight percent of B-HEMA or ETPTA. The PVDF film and Celgard were also immersed in the liquid electrolyte to compare electrochemical properties with L-GPEs and H-GPEs, and the resulting electrolytes are named as e-PVDF and e-Celgard. Scheme 4 shows the compositions of the electrolytes.

Characterization. ^1H NMR spectra were recorded on an Ascend 400 spectrometer (300 MHz for ^1H NMR) and ^{11}B NMR spectra were recorded on a JeolJNM-LA400 spectrometer (400 MHz) with LFG. CDCl_3 (Cambridge Isotope Laboratories) was used as a solvent at room temperature with TMS as a reference. The Fourier transform infrared (FT-IR) spectra were recorded in the absorption mode on Nicolet 6700 spectrophotometer with a resolution of 4 cm^{-1} in the vibrational frequency range from 400 to 4000 cm^{-1} . Field-emission scanning electron microscopy (FE-SEM) was performed on a JEOL JSM-6700F with an accelerating voltage of 10 kV. The X-ray diffraction (XRD) spectra were obtained using Rigaku SmartLab (Cu $K\alpha$) spectrometers. The liquid electrolyte contact angles were measured at room temperature and ambient relative humidity using a Krüss DSA 10 contact angle analyzer connected to drop shape analysis software. Raman spectra were collected on a T64000 Triple Raman spectrometer (HORIBA) equipped with a 514.5 nm Ar laser source. The contact angles for each sample were measured more than five times on five independently prepared samples, and the average values were used. The mechanical properties were measured using a universal testing machine (LS1SC, LLOYD Instruments). The dumbbell specimens were prepared using the ASTM standard D638 (Type V specimens dog-bone shaped samples). The tensile properties of the membrane samples were measured with a gauge length and cross head speed of 15 mm and 5 mm/min, respectively. At least five specimens for each sample were tested, and their average values were used. The amount of liquid electrolyte uptake was determined by measuring their changes in weight before and after immersion in the liquid electrolyte for 24 h. The electrolyte uptake value is calculated as the following eq 1:

$$\text{electrolyte uptake (\%)} = [(W_{\text{wet}} - W_{\text{dry}})/W_{\text{dry}}] \times 100 \quad (1)$$

where W_{dry} is the weights of the dry films of PVDF, Celgard, or IPMs and W_{wet} is the weights of e-PVDF, e-Celgard, or GPEs.

Electrochemical Characterization. The ionic conductivities of the e-PVDF, e-Celgard, and GPEs were determined using complex impedance spectroscopy with a Zahner Elektrik IM6 apparatus in the frequency range of 0.1 Hz to 1.0 MHz with 10 mV of AC amplitude. The samples for the measurements were prepared by sandwiching the e-PVDF, e-Celgard, and GPEs between two stainless-steel electrodes in 2032 coin cell. The ionic conductivity (σ) was calculated from the following eq 2:

$$\sigma = (1/R) \times (d/A) \quad (2)$$

where R is electrolyte resistance obtained from the impedance spectrum, d is the thickness of electrolyte, and A is the area of the electrode. Lithium transference number (t_{Li^+}) was determined by DC polarization/AC impedance combination method.²⁹ The GPEs were sandwiched between two nonblocking lithium metal disks to form a symmetrical Li/GPE/Li in 2032 coin cell. The cell was polarized by a constant DC voltage of 10 mV and following current values were monitored until steady-state current was observed. The initial and steady-state resistances of the cell were also measured. From this method, t_{Li^+} was determined by following eq 3:

$$t_{\text{Li}^+} = \frac{I_s(V - I_i R_i)}{I_i(V - I_s R_s)} \quad (3)$$

where V is the constant DC voltage applied to the cell and R_i and R_s are the initial and steady-state resistances, respectively. I_i and I_s are the initial and steady-state currents, respectively. This test was also performed using e-PVDF and e-Celgard as the electrolytes. Electrochemical stability was evaluated by linear sweep voltammetry (LSV) and cyclic voltammetry (CV) using a potentiostat (VMP3, Biologics) at 30°C at a scan rate of 1 mV/s. The cell for the measurement was assembled by sandwiching GPE between stainless steel (working electrode) and lithium metal (reference electrode) in 2032 coin cell. The cell was swept in a potential range from 3 to 7 V (versus Li/Li $^+$) at a scan rate of 1 mV/s at 30°C . Charge/discharge test of lithium-ion battery was performed with WBCS3000 battery cycler (WonATech) at 30°C . LiV_3O_8 (70 wt %) prepared as previously described⁴² was used as cathode active materials and dispersed in NMP with carbon black (20 wt %) and PVDF (10 wt %). The resultant slurry was deposited and cast onto Aluminum current collector using a doctor blade. The residual NMP was completely removed under vacuum condition at 120°C for 24 h. The obtained cathode sheet, lithium metal, and GPE

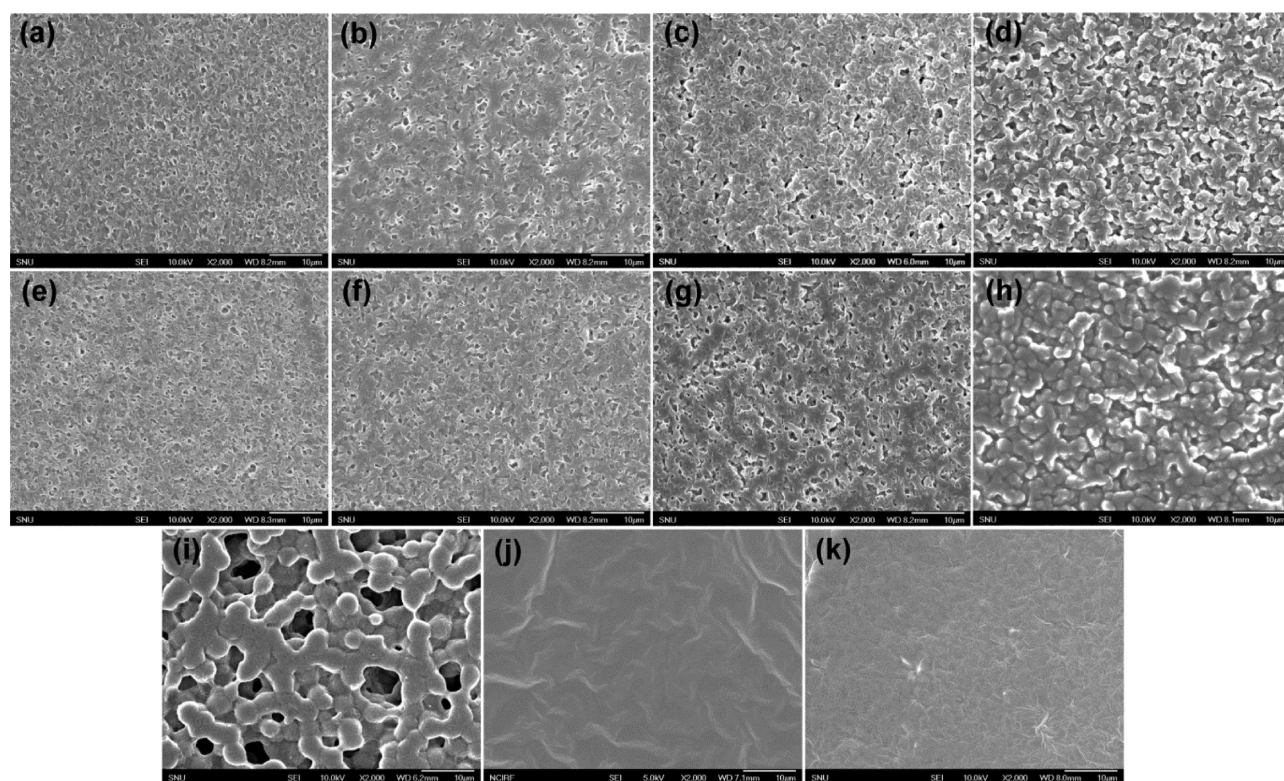


Figure 2. SEM images of (a) L-IPM20, (b) L-IPM40, (c) L-IPM60, (d) L-IPM80, (e) H-IPM20, (f) H-IPM40, (g) H-IPM60, (h) H-IPM80, (i) PVDF film, (j) UV-cured LBC film, and (k) UV-cured HBC film (scale bar = 10 μm), where the number after each IPM represents the weight percent of LBC or HBC.

were punched into disk shapes and assembled together in 2032 coin cell to form Li/GPE/LiV₃O₈ cell. All components were assembled in argon filled glovebox ($\text{H}_2\text{O} < 0.5$ ppm, $\text{O}_2 < 0.5$ ppm). Charge/discharge test of the lithium-ion battery was performed at cutoff voltages of 2.0–3.6 V versus Li/Li⁺ at 30 °C with a C-rate of 1.0 C, where 1.0 C rate corresponds to a current density of 280 mA g^{−1}. This test was also performed using e-PVDF and e-Celgard as the electrolytes. Rate capability of Li/L-GPE80/LiV₃O₈ cell was tested at the same cutoff voltages at 30 °C by changing various current densities.

RESULTS AND DISCUSSION

Synthesis and Characterization of Boron-Containing Cross-Linkers (BCs). Boron-containing cross-linkers (BCs) such as LBC and HBC having EO chains, cross-linkable methacrylate groups, and anion-trapping boron moieties were synthesized by the substitution reaction of TMB and PEGMA as shown in Scheme 2a,b. B-HEMA was also prepared from the reaction of TMB and HEMA (Scheme 2c). The reaction at 50 °C for 3 h reached the equilibrium state and further 2 h at 70 °C under nitrogen gas blowing removed the byproduct, methanol, to produce BCs with high yields. PEGMAs with different molecular weight (average $M_n = 360$ g mol^{−1} and 500 g mol^{−1}) were used as the reactants to study the effect of the EO chain length on various electrochemical properties of the GPEs. Structure of the LBC and HBC was confirmed by ¹H NMR analysis (Figure 1a,b). Signals a and e are assigned to the protons of the methacrylate group, and signals b and d are attributed to the protons of EO units. The peaks in the range of 3.88–4.05 ppm (signal c) represent the presence of protons in the EO units nearby borate (B–O–CH₂–), verifying the successful incorporation of the borate into the PEGMA.³⁷ Furthermore, when the hydroxyl groups of PEGMA are

substituted by the borate, the proton peaks of EO chain-end shifted downfield because of electron deficient nature of the borate. The conversion value corresponding to the fraction of PEGMA linked to the borate by the reaction was calculated by the following equation:

$$\text{conversion (\%)} = (I_c/2)/(I_e/3) \times 100 \quad (4)$$

where I_c and I_e are the intensities of the peaks c and e, respectively. The conversion values of LBC and HBC are very close such as 79% and 77%, respectively, indicating that the reactivity of both PEGMA having different molecular weights is very close. The BCs were further analyzed by ¹¹B NMR as shown in Figure 1c. TMB showed a sharp boron peak at 17.5 ppm, whereas both LBC and HBC exhibited very broad peaks at 17.4 ppm corresponding to the characteristic signal of the tricoordinated boron atoms. The slight peak shift and broadness of the peaks are ascribed to the attached EO chains.

Preparation and Properties of Semi-IPN Polymer Membranes (IPMs). A series of semi-IPN polymer membranes (IPMs) having different weight ratios of BC and PVDF such as 20:80, 40:60, 60:40, and 80:20 were prepared by the solution casting method followed by the UV curing process to cross-link the methacrylate groups of BC as shown in Scheme 3 and 4. The IPMs prepared from LBC and HBC are abbreviated as L-IPMs and H-IPMs, respectively, and the number after each abbreviation represents the weight percentage of BCs in the IPMs. The IPMs containing different BC content (20, 40, 60, and 80 wt %) are all flexible and free-standing films as shown in Figure S1 in the Supporting Information. The maximum BC content was found to be about 80 wt %, because IPMs having BC content larger than 80 wt % cannot maintain their dimensionally stable free-standing film state after the

immersion in the liquid electrolyte. Although BCs alone can be cross-linked without PVDF by UV irradiation forming cross-linked films as shown in Figure S2(a) and (b) (Supporting Information), these films did not maintain the pristine dimensional stability without PVDF after they were immersed in the liquid electrolyte because EO units in LBC and HBC uptake the liquid electrolyte too much. As shown in Figure S2(c), electrolyte uptake values of both UV-cured LBC and HBC films increase up to about 400% after immersion in liquid electrolyte for 6 h, and then they severely swell up and do not maintain the free-standing film state, indicating that they are not dimensionally stable enough to be used as GPEs. The cross-linking reactions were confirmed by the FT-IR analysis (Figure S3 in the Supporting Information). The peaks at 1637 cm^{-1} assigned to the characteristic vinyl ($\text{C}=\text{C}$) stretching from both LBC and HBC clearly disappeared after the UV curing process for 5 min, indicating that the methacrylate groups of BCs produced the cross-linked networks.^{16,43}

The morphological properties of the IPMs, the PVDF film, and the UV-cured LBC and HBC films were investigated by SEM as shown in Figure 2. Although the porous structure with large pore size and porosity in GPE is beneficial to increase the ionic conductivity owing to the increased liquid electrolyte uptake ability,⁴⁴ the pore size in the GPE should not be too much large because pores larger than $1\text{ }\mu\text{m}$ can cause the leakage of electrolytes or active materials and also the penetration of lithium dendrites.^{18,45–47} It was found that the PVDF film (Figure 2i) prepared from the same procedure used for the IPM preparation has pores having diameters larger than $5\text{ }\mu\text{m}$, whereas the UV-cured LBC (Figure 2j) and HBC films (Figure 2k) show dense and wrinkled surfaces without any pores. L-IPMs and H-IPMs show interconnected porous structures with pores smaller than $1\text{ }\mu\text{m}$, i. e., the maximum pore size showing high ionic conductivity without any leakage and penetration problems.

Figure 3 shows the XRD patterns of H-IPM80, L-IPM80, and the PVDF film. The PVDF film shows three peaks at $2\theta =$

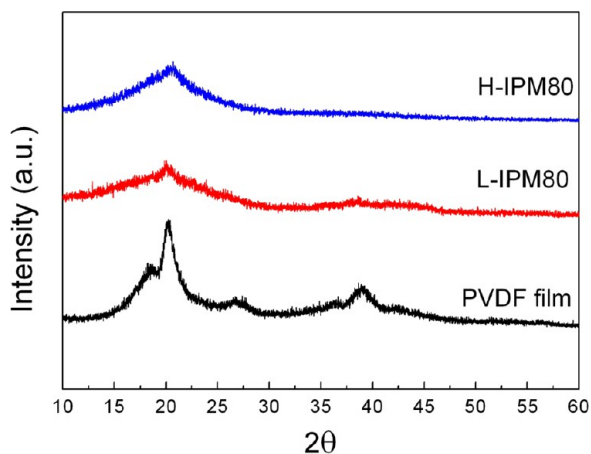


Figure 3. XRD patterns of H-IPM80, L-IPM80, and PVDF film.

18.2° , 20.0° , and 26.6° corresponding to the (100), (020), and (110) reflections for the α -phase crystalline structure.⁴⁸ Such crystalline peaks are not clearly observed from L-IPM80 and H-IPM80, because the introduction of LBC and HBC can prevent the crystallization of PVDF chains in the IPMs. The XRD patterns of L-IPMs and H-IPMs having various BC contents in Figure S4 in the Supporting Information show that the

intensities of the crystalline peaks decrease with increasing BC content. PVDF in amorphous phase is known to absorb liquid electrolyte, and then it can be used as GPE.²⁶ Therefore, both the amorphous PVDF and cross-linked BC networks having EO chains in the IPMs can uptake the electrolyte, which in turn can increase the ionic conductivity.

Anion-Trapping Behavior of Lewis Acidic Boron Moieties. Anion-trapping ability of boron moieties in the GPEs was evaluated by ^{11}B NMR analysis. As shown in Figure 4, L-IPM80 has a single broad peak at about 18 ppm

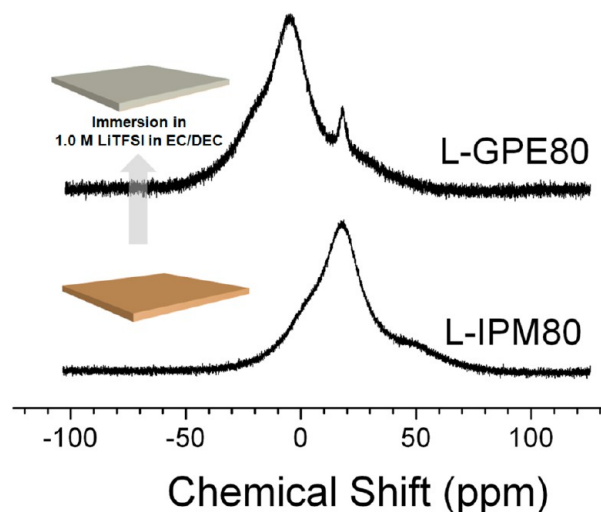


Figure 4. Solid-state ^{11}B NMR spectra of L-GPE80 and L-IPM80.

corresponding to characteristic signal of tricoordinated boron which matches well with that of LBC as presented in Figure 1(d). L-GPE80 obtained by immersing L-IPM80 in 1.0 M LiTFSI in EC:DEC (1:1 vol %) for 24 h show a new boron peak at about -5 ppm from the interaction between boron atom and TFSI $^-$ anion in the liquid electrolyte, indicating that boron traps the anion of the lithium salt.^{40,65} Raman spectra of E-GPE80 (cross-linker: ETPTA) and L-GPE80 (cross-linker: LBC) were measured to further confirm the anion-trapping ability of boron moiety (Figure 5) by comparing two bands at $742\text{ and }748\text{ cm}^{-1}$ corresponding to $\delta_s(\text{CF}_3)$ of lithium-uncoordinated TFSI $^-$ and lithium-coordinated TFSI $^-$, respectively.^{66,67} It was found that the intensity of lithium-uncoordinated TFSI $^-$ band at 742 cm^{-1} in L-GPE80 having the anion-trapping boron moiety is much larger than that in E-GPE80 without any boron moiety.

Ion Transport Properties of Gel Polymer Electrolytes (GPEs). For the electrochemical studies, e-PVDF, e-Celgard, and GPEs were prepared by immersing the PVDF film, Celgard, and IPMs in 1.0 M LiTFSI in EC:DEC (1:1 vol %) for 24 h (Scheme 3). Figure 6 shows the electrolyte uptake behavior (blue dashed lines) of the e-PVDF, e-Celgard, and GPEs with various BC contents. The electrolyte uptake value of the PVDF film is smaller than IPMs due to the presence of crystalline phase in the PVDF. Because the increase of BC content in GPEs can increase the content of EO chains and also decrease the crystallinity of PVDF, GPEs with larger BC content show larger electrolyte uptake values. Celgard, the commercialized polymer separator, exhibits 91% of electrolyte uptake that is larger than those of L-IPM20 and H-IPM20 having the smallest BC content, indicating that 20 wt % of BC is not sufficient to absorb enough amount of the electrolyte.

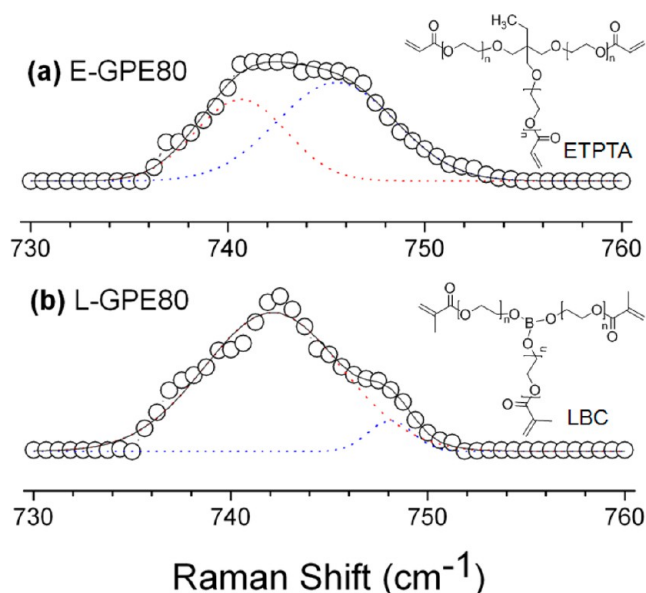


Figure 5. Raman spectra of (a) E-GPE80 and (b) L-GPE80.

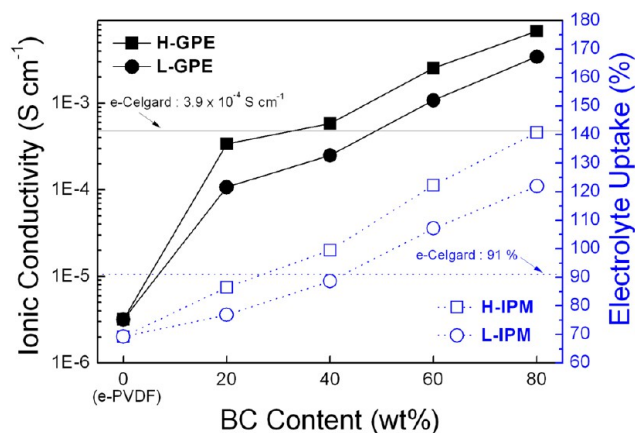


Figure 6. Ionic conductivities (black solid lines) of H-GPEs and L-GPEs and electrolyte uptake values (blue dashed lines) of H-IPMs and L-IPMs having various BC contents at 30 °C. e-PVDF and e-Celgard are the electrolytes prepared by immersing PVDF film and Celgard into the 1.0 M LiTFSI in EC:DEC (1:1 vol %), respectively.

However, both L-IPMs and H-IPMs having BC content larger than 40 wt % exhibit larger electrolyte uptake values than Celgard, because larger contents of EO chains can increase the electrolyte uptake due to the polar affinity toward the liquid electrolyte. It was also found that the electrolyte uptake values of H-IPMs are always larger than those of L-IPMs at all the BC contents, because H-IPMs contain larger amount of EO chains. H-IPM80 having largest amount of EO chains exhibited 141% of the electrolyte uptake value that is 72% and 50% larger than those of the PVDF film and Celgard, respectively.

To evaluate the electrolyte wettability of the IPMs, contact angles of liquid electrolyte droplet on Celgard, L-IPM80, and H-IPM80 were measured (Figure S5 in the Supporting Information). Celgard comprising hydrophobic polyolefin exhibited the largest contact angle value of 52.6°, whereas L-IPM80 and H-IPM80 having hydrophilic EO chains exhibited much smaller contact angle values of 16.2° and 15.0°, respectively, indicating that the IPMs are more wettable to the liquid electrolyte. Notably, H-IPM80 having a larger

amount of hydrophilic EO chains shows smaller contact angle value than L-IPM80.

Ionic conductivities (black solid lines) of L-GPEs and H-GPEs as a function of BC content at 30 °C are also shown in Figure 6. The ionic conductivities of both L-GPEs and H-GPEs increase with increasing BC content as expected from the electrolyte uptake behavior. When the BC content reaches 60 wt %, the ionic conductivity values of both H-GPEs and L-GPEs are larger than that of e-Celgard. Maximum ionic conductivity was obtained from H-GPE80 ($4.2 \times 10^{-3} \text{ S cm}^{-1}$) and L-GPE80 ($3.4 \times 10^{-3} \text{ S cm}^{-1}$), and these values are 1 order of magnitude larger than that of the e-Celgard ($3.9 \times 10^{-4} \text{ S cm}^{-1}$) at 30 °C. Meanwhile, H-GPEs having larger electrolyte uptake values show larger ionic conductivity than L-GPEs having smaller electrolyte uptake values. The larger ionic conductivity of H-GPEs might be also affected by the cross-linking density.⁴⁹ Because H-GPE and L-GPE are prepared using the same weight percent of BC, H-GPE should have less amount of cross-linkable methacrylate group, because HBC used for the preparation of H-GPE (or H-IPM) have higher molecular weight than LBC for L-GPE. The cross-linking density in GPEs could be estimated by observing the glass transition temperatures (T_g 's) of IPMs. As shown in the DSC curves in Figure S6 (Supporting Information), the T_g 's of L-IPM80 and H-IPM80 are -41.4 and -48.9 °C, respectively; smaller T_g indicates less cross-linking density for the polymer systems with the same chemical compositions, and the polymer with smaller cross-linking density can have more chain mobility to increase the ionic conductivity.⁴⁹ Furthermore, temperature-dependence of ionic conductivities of L-GPE80, H-GPE80, and e-Celgard was also measured (Figure S7 in the Supporting Information).

The anion-trapping property of Lewis acidic boron moieties was quantitatively evaluated by measuring the lithium transference number (t_{Li^+}) using DC polarization/AC impedance combination method. Chronoamperometric curves and electrochemical impedance spectra of L-GPE80, H-GPE80, BH-GPE80, and e-Celgard are presented in Figure S8 in the Supporting Information. Figure 7 shows the t_{Li^+} values of H-GPEs and L-GPEs having various BC contents. The t_{Li^+} values

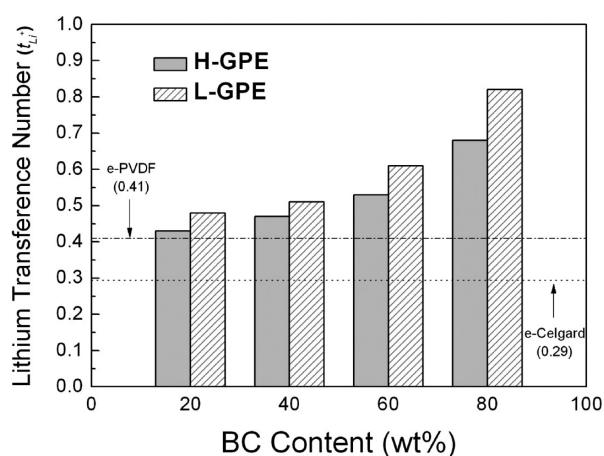
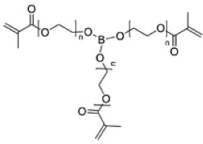
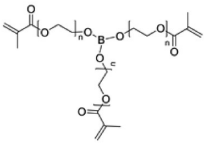
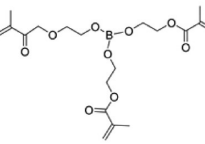
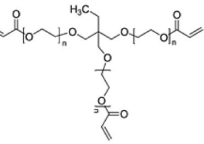


Figure 7. Lithium transference numbers (t_{Li^+} 's) of e-PVDF, e-Celgard, and H-GPEs/L-GPEs having various BC contents at 30 °C, where e-PVDF and e-Celgard are the electrolytes prepared by immersing PVDF film and Celgard into the 1.0 M LiTFSI in EC:DEC (1:1 vol %), respectively.

Table 1. Structure of Crosslinkers, EO and Boron Content, Ionic Conductivity, and Lithium Transference Number (t_{Li+}) of L-GPE80, H-GPE80, BH-GPE20, and E-GPE80, where the Number after GPE Represents the Weight Percent of Each Crosslinker^a

	L-GPE80	H-GPE80	BH-GPE20	E-GPE80
Crosslinker	 LBC	 HBC	 B-HEMA	 ETPTA
# of EO unit (n)	6.2	9.4	1	4.7
EO content (mol/ 1 g of IPM)	1.3×10^{-2}	1.5×10^{-2}	1.5×10^{-3}	1.2×10^{-2}
Boron content (mol/ 1 g of IPM)	5.8×10^{-4}	4.1×10^{-4}	4.4×10^{-4}	0
Ionic conductivity (σ , S cm ⁻¹)	3.4×10^{-3}	4.2×10^{-3}	1.6×10^{-5}	1.3×10^{-3}
t_{Li+}	0.82	0.68	0.70	0.30

^ae-PVDF: $\sigma = 3.2 \times 10^{-6}$ S cm⁻¹, $t_{Li+} = 0.41$. e-Celgard: $\sigma = 3.9 \times 10^{-4}$ S cm⁻¹, $t_{Li+} = 0.29$

of e-PVDF (0.41) and e-Celgard (0.29) are always smaller than those of all the H-GPEs and L-GPEs, because the boron moieties in the BC trap the anion of the lithium salt. The increase in the BC content in GPEs increases the t_{Li+} value, and those of L-GPEs are always larger than those of H-GPEs when they have the same BC content. Obviously, the GPEs having a larger amount of Lewis acidic boron moiety show the larger t_{Li+} value. For GPEs prepared using the same amount of BCs, L-GPE contains larger content of boron moiety than H-GPE, because the molecular weight of LBC is smaller than that of HBC. As a result, L-GPE80 having the largest content of boron moiety shows the largest t_{Li+} value of 0.82.

BH-GPE20 was prepared from B-HEMA to investigate the effect of EO chains on the electrochemical properties by comparing with H-GPE80, because the boron content of BH-GPE20 (4.4×10^{-4} mol) is very close to that of H-GPE80 (4.1×10^{-4} mol), whereas the EO content of BH-GPE20 (1.5×10^{-3} mol) is much smaller than that of H-GPE80 (1.5×10^{-2} mol) (Table 1). The t_{Li+} value of BH-GPE20 (0.70) was found to be close to that of H-GPE80 (0.68) because they have similar boron contents. In contrast, the ionic conductivity values of BH-GPE20 (1.6×10^{-5} S cm⁻¹) is much smaller than that of H-GPE80 (4.2×10^{-3} S cm⁻¹) because BH-GPE20 contains a much smaller amount of EO chains than H-GPE80, indicating that the EO chains significantly contribute to increase ionic conductivity of the GPEs, because EO is the lithium-ion conducting and electrolyte uptake unit. For example, the electrolyte uptake values of H-IPM80 and BH-IPM20 are 141% and 60%, respectively.

E-GPE80 was also prepared from well-known commercialized cross-linker, ETPTA (average $M_n = 912$ g mol⁻¹) to investigate the effect of boron because it does not have any boron moiety (the boron content is 0%), whereas the EO

contents of E-GPE80 (1.2×10^{-2} mol) is quite close to that of L-GPE80 (1.3×10^{-2} mol) (Table 1). Although their EO contents are similar, the ionic conductivity of E-GPE80 (1.3×10^{-3} S cm⁻¹) without any boron moiety is quite smaller than that of L-GPE80 (3.4×10^{-3} S cm⁻¹) having the boron moiety because electrolyte uptake value of E-GPE80 (100%) is smaller than that of L-GPE80 (122%). It is also well-known that the incorporation of boron moieties can enhance the lithium-ion transport property including t_{Li+} value;¹⁰ the t_{Li+} value of L-GPE80 (0.82) is 2.7 times larger than that of E-GPE80 (0.30). The increase of t_{Li+} values by the boron moiety was further investigated using solid polymer electrolytes prepared using only cross-linkers (LBC and ETPTA) without any PVDF and liquid electrolyte (Figure S9 in the Supporting Information). It was found that t_{Li+} value of UV-cured LBC (0.62) is larger than that of UV-cured ETPTA (0.24) due to the anion-trapping ability of the boron moiety in the LBC. Although the t_{Li+} values of these solid polymer electrolytes are quite smaller than those of GPEs (L-GPE80:0.82 and E-GPE80:0.30) due to the different kinetics arisen from the solid and gel natures, respectively, the same anion-trapping ability of the boron moiety could be further confirmed.

Mechanical, Thermal, and Electrochemical Stabilities. Mechanical properties of L-IPM80, H-IPM80, Celgard, and PVDF film were evaluated using universals testing machine (Figure S10 in the Supporting Information). It was found that L-IPM80 exhibits larger tensile strength and Young's modulus values than H-IPM80, whereas the percentage strain at break value of L-IPM80 is smaller than that of H-IPM80 because L-IPM80 has larger cross-linking density than H-IPM80 as described above. The mechanical strength of L-GPE80 and H-GPE80 could not be measured because these films became very fragile after the liquid electrolyte immersion process. Although

the large electrolyte uptake ability of L-GPE80 and H-GPE80 containing large amount of EO chains can enhance the electrochemical performances, it can decrease the mechanical strength of the GPEs. It was also found that the mechanical strength of L-IPM80 and H-IPM80 are much smaller than those of Celgard and PVDF film (Figure S10(b)). Therefore, large-scale production using the current cell assembly process might not be possible using the boron-containing polymer systems in this study.⁶⁹ Further works should be needed to prepare the lithium batteries using the boron-containing polymers in the industrial process.

A thermal shrinkage test was conducted on Celgard, the PVDF film, L-IPM80, and H-IPM80 as shown in Figure 8.

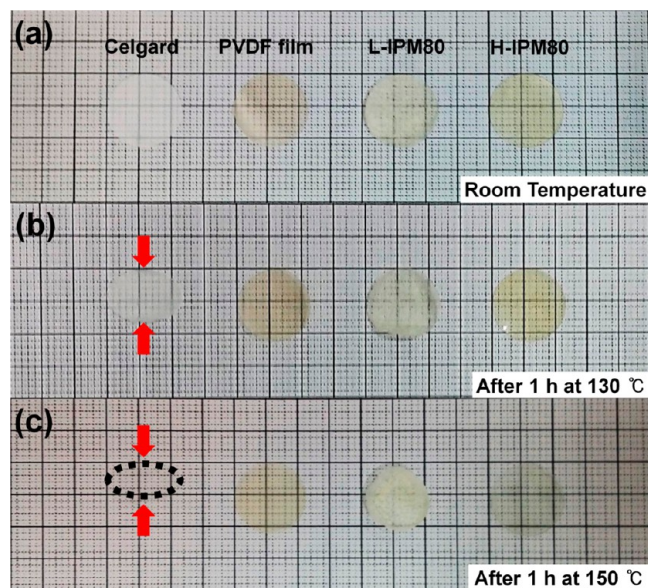


Figure 8. Thermal shrinkage behaviors of Celgard, PVDF film, L-IPM80, and H-IPM80 (a) at room temperature, (b) after 1 h at 130 °C, and (c) after 1 h at 150 °C.

Because L-IPM80 and H-IPM80 exhibit largest ionic conductivities among L-IPMs and H-IPMs, respectively, they were used for further studies. It was found that Celgard shrank uniaxially after heating at 130 °C for 1 h due to the internal stress originated during the stretching process.^{50,51} Celgard became transparent after heating at 150 °C for 1 h, indicating the transition from the crystalline to amorphous phase.⁵² The thermal shrinkage behavior of conventional olefin based separator causes internal short-circuit failure of lithium-ion batteries.^{53,54} In contrast, such thermal shrinkage behavior was not observed up to 150 °C from L-IPM80 and H-IPM80 due to the presence of thermally stable PVDF matrix, indicating that our IPMs are more suitable for the high-temperature applications than the commercialized separator systems. Furthermore, cross-linked IPM structures also contribute to the improvement of thermal stability.²⁸ Combustion test was also conducted on Celgard, the PVDF film, L-IPM80, and H-IPM80 as shown in Figure S11 in the Supporting Information. When Celgard was contacted with fire, it shrank immediately and was set on fire within few seconds, possibly due to the combustible nature of polyolefin matrix. In contrast, the PVDF film maintained its original shape after contact with fire and was not set on fire because of the flame retarding and self-extinguishing properties of PVDF. The combustion test results

of L-IPM80 and H-IPM80 were close to that of the PVDF film, indicating that our membranes exhibit better thermal stability than Celgard.

Electrochemical stability of L-GPE80 and H-GPE80 was evaluated by linear sweep voltammetry (LSV) (Figure 9a). The

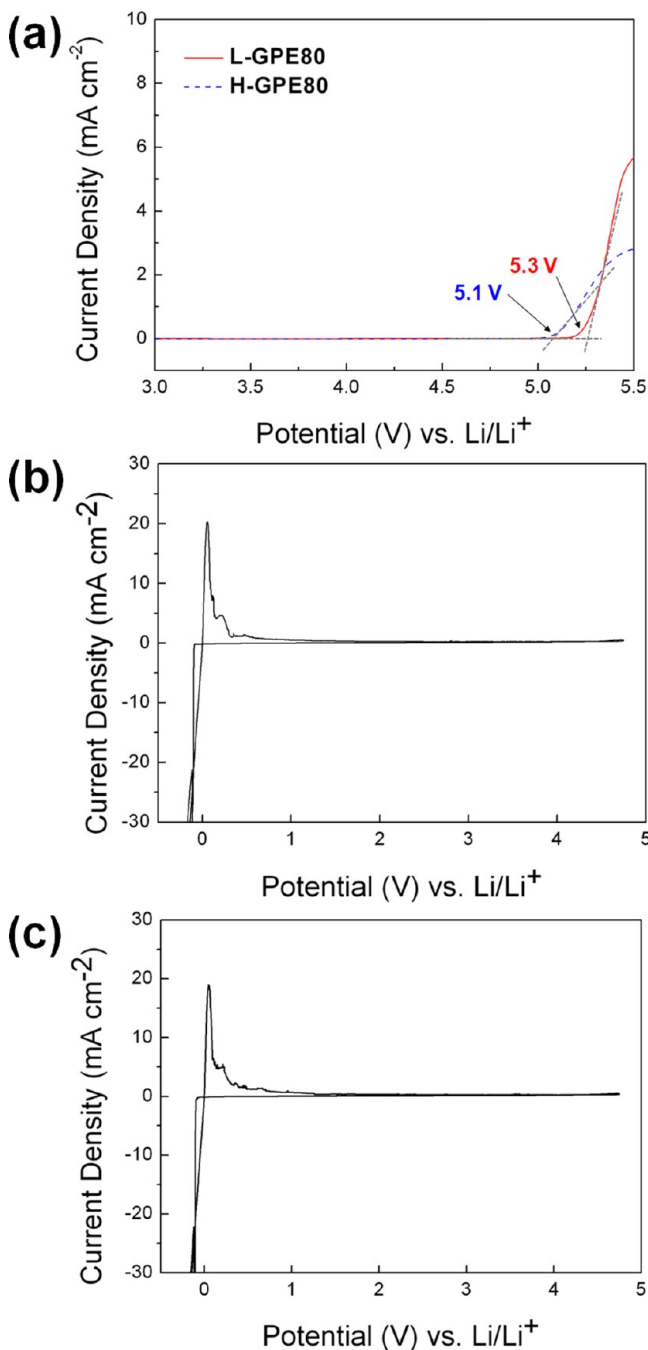


Figure 9. (a) Linear sweep voltammogram of L-GPE80 and H-GPE80 and cyclic voltammogram of (b) L-GPE80 and (c) H-GPE80 at 30 °C with a scan rate of 1 mV/s.

abrupt increase in the current corresponds to the electrochemical decomposition of anions of the lithium salt in the liquid electrolyte. The decomposition voltage values of L-GPE80 and H-GPE80 were found to be 5.3 and 5.1 V (vs Li/Li⁺), respectively. The excellent electrochemical stability could be attributed to strong complexation between the boron and

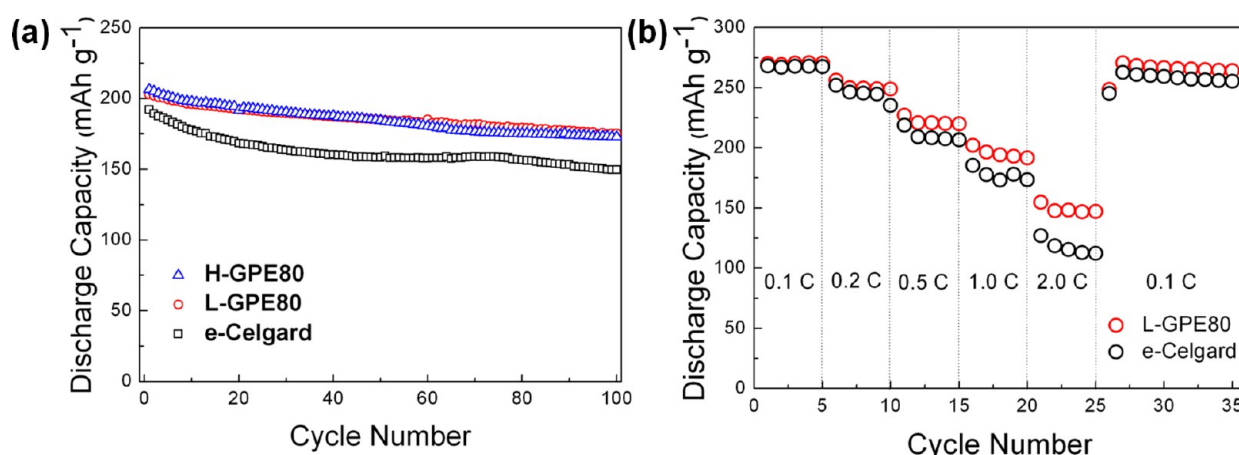


Figure 10. (a) Discharge capacity profiles of Li/electrolytes/LiV₃O₈ cells cycled at 30 °C with a C-rate of 1.0 C, where the electrolytes are H-GPE80, L-GPE80, and e-Celgard and (b) rate property of L-GPE80 and e-Celgard at 30 °C with various C-rates.

TFSI anions of the lithium salt, resulting in the retardation of decomposition of TFSI anions.^{37,39} Furthermore, it was found that electrochemically stable window of L-GPE80 is wider than that of H-GPE80, because L-GPE80 contains larger boron content than H-GPE80. It can be concluded that the our GPEs having boron moieties are electrochemically stable up to 5.0 V within the operating voltage range of the LiV₃O₈ cathode as well as high-voltage cathode materials. It was also found that our GPE system is compatible with lithium metal anode because reversible lithium deposition/dissolution behavior at around 0 V (vs Li/Li⁺) in the cathodic cyclic voltammetry scan was observed from both L-GPE80 and H-GPE80 (Figure 9b,c).

Cycle Performance. Cycle performance of the batteries assembled with Li/electrolytes/LiV₃O₈, where the electrolytes are L-GPE80, H-GPE80, and e-Celgard was evaluated at 30 °C at a C-rate of 1.0 C for 100 cycles. Since lithium metal having very large specific capacity (~3860 mAh g⁻¹) was used as the anode materials, it is desirable to use the cathode materials with specific capacity as large as possible. The specific capacity of LiV₃O₈ (280 mAh g⁻¹) is larger than those of the other commonly used active materials such as LiCoO₂ (140 mAh g⁻¹) and LiFePO₄ (170 mAh g⁻¹), although the dissolution of vanadium in the LiV₃O₈ system has been known to be disadvantageous.^{55,56} Comparing the initial discharge capacity values of the electrolyte system, H-GPE80 having the highest ionic conductivity shows the largest initial discharge capacity (Figure 10a). For the cycle life behavior, the GPEs show larger capacity retention values (87% for L-GPE80 and 83% for H-GPE80) after 100 cycles than e-Celgard (78%). The better capacity retention behavior in the GPEs could be ascribed to the suppressed interfacial side reactions between the liquid electrolyte and the electrode surface, because the interaction of the liquid electrolytes with polymer matrix containing EO units in the GPE is much stronger than that in the e-Celgard system as reported by others.^{57,58} The different cyclability of L-GPE80, H-GPE80, and e-Celgard was further studied by electrochemical impedance spectroscopy. Figure S12 in the [Supporting Information](#) shows electrochemical impedance spectra of L-GPE80, H-GPE80, and e-Celgard before and after cycling at 1.0 C for 100 cycles. L-GPE80 and H-GPE80 show relatively smaller increase in the interfacial resistance after 100 cycles (L-GPE80: from 17.7 to 54.0 ohm and H-GPE80: 18.0 to 65.1 ohm) than e-Celgard (from 23.4 to 202 ohm), because further growth of resistive solid electrolyte interphase (SEI) layer on

the electrode during cycling is significantly suppressed with the incorporation of GPE.⁶⁸ Meanwhile, L-GPE80 exhibited larger cycle retention value than H-GPE80 possibly because the boron content in L-GPE80 is larger than that in H-GPE80. It was reported that the anion-trapping boron moieties can reduce the concentration polarization by localized anions which in turn can improve the capacity retention even at a high C-rate such as 1.0 C.⁵⁹ Therefore, although L-GPE80 has the smaller ionic conductivity value than H-GPE80, the larger boron concentration in L-GPE80 can improve the capacity retention.

Figure 10b shows the discharge capacity profiles obtained by increasing the C-rate from 0.1 to 2.0 C for every five cycles using the battery assembled with e-Celgard and L-GPE80 exhibiting the best cycle performance. The discharge capacity values of both L-GPE80 and e-Celgard are very close to the theoretical capacity value of LiV₃O₈ (280 mAh g⁻¹) at a scan rate of 0.1 C. Although the discharge capacity value gradually decreases as the C-rate increases from 0.1 to 2.0 C, L-GPE80 shows relatively larger capacity values than e-Celgard even at a high C-rate like 2.0 C, indicating that the L-GPE80 has better rate property possibly due to the presence of anion-trapping boron moieties and relatively large ionic conductivity. Furthermore, the discharge capacity value recovers up to the initial values when the C-rate is decreased from 2.0 to 0.1 C again, indicating the reversible capacity recovery behavior of L-GPE80. Galvanostatic charge/discharge profiles of L-GPE80 and e-Celgard cycled at various C-rates are presented in Figure S13 in the [Supporting Information](#). It was found that the voltage difference between the discharge–charge plateaus increases with increasing C-rate due to electrode polarization at high current density. Furthermore, discharge plateau becomes steeper and shorter as the C-rate increases, resulting in decreased capacity value. As shown in Figure S13(c–e), L-GPE80 exhibits higher plateau voltage and longer plateau range than e-Celgard, resulting in larger capacity value, indicating that electrode polarization is significantly reduced when the anion-trapping boron moiety is incorporated.

CONCLUSIONS

In this study, GPEs based on semi-IPN structure containing PVDF and cross-linkers having anion-trapping boron moieties and EO chains were prepared for lithium-ion battery applications. Dimensional stability of the GPE is maintained even at elevated temperature up to 150 °C due to presence of

thermally stable PVDF matrix. The maximum ionic conductivity of 4.2 m S cm^{-1} at 30°C was obtained from H-GPE80, and this value is 1 order of magnitude larger than that of the liquid electrolyte containing commercialized Celgard separator due to large electrolyte uptake ascribed from the EO chains. A large lithium transference number (t_{Li^+}) of 0.82 was obtained from L-GPE80 because of the introduction of a large amount of anion-trapping boron moieties. Furthermore, good electrochemical stability up to 5.4 V was obtained because strong complexation between the boron moieties and anions of the lithium salt can retard the electrochemical oxidative decomposition of the electrolyte. All these electrochemical advantages of our GPEs contribute to achieve excellent cycle performance at a high C-rate of 1.0 C during 100 cycles.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acsami.6b09601](https://doi.org/10.1021/acsami.6b09601).

Additional figures: Photographs of IPMs (Figure S1), photographs and time-dependent electrolyte uptake values of UV-cured LBC and HBC films (Figure S2), FT-IR spectra of HBC, H-IPM80, LBC, and L-IPM80 (Figure S3), XRD patterns of PVDF film, L-IPMs, H-IPMs having various BC contents (Figure S4), liquid electrolyte contact angles of Celgard, L-IPM80, and H-IPM80 (Figure S5), DSC curves of L-IPM80 and H-IPM80 (Figure S6), temperature-dependence of ionic conductivities of L-GPE80, H-GPE80, and e-Celgard (Figure S7), chronoamperometric curves and EIS spectra of L-GPE80, H-GPE80, BH-GPE80, and e-Celgard (Figure S8), chronoamperometric curves and EIS spectra of UV-cured LBC having LiTFSI and UV-cured ETPTA having LiTFSI (Figure S9), stress-strain curves of L-IPM80, H-IPM80, Celgard, and PVDF (Figure S10), combustion test on Celgard, PVDF film, L-IPM80, and H-IPM80 (Figure S11), electrochemical impedance spectra before and after cycling at 1.0 C for 100 cycles (Figure S12), and voltage-capacity curves of L-GPE80 and e-Celgard at various C-rates (Figure S13). (PDF)

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Notes

The authors declare no competing financial interest.

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